

Appendix Figures

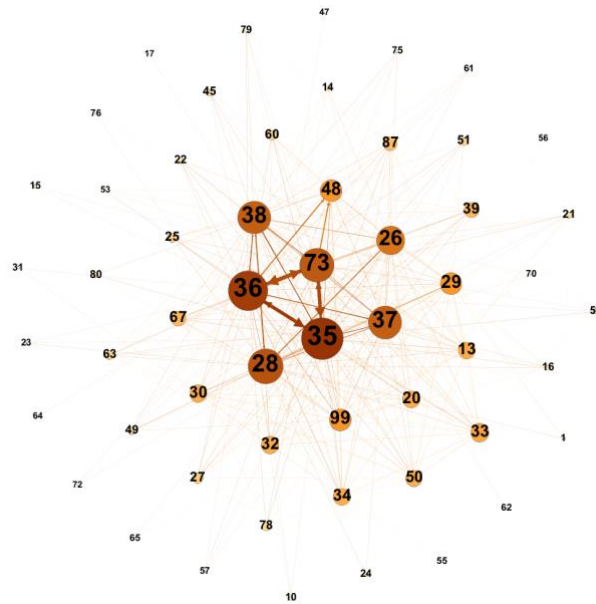


Figure 1: Citation network corresponding to year 1997; each node represents an industry; industries are identified by their 2-digit SIC codes; the size of a node is proportional to its out-degree; the direction of the edge is from the cited to the citing industry.

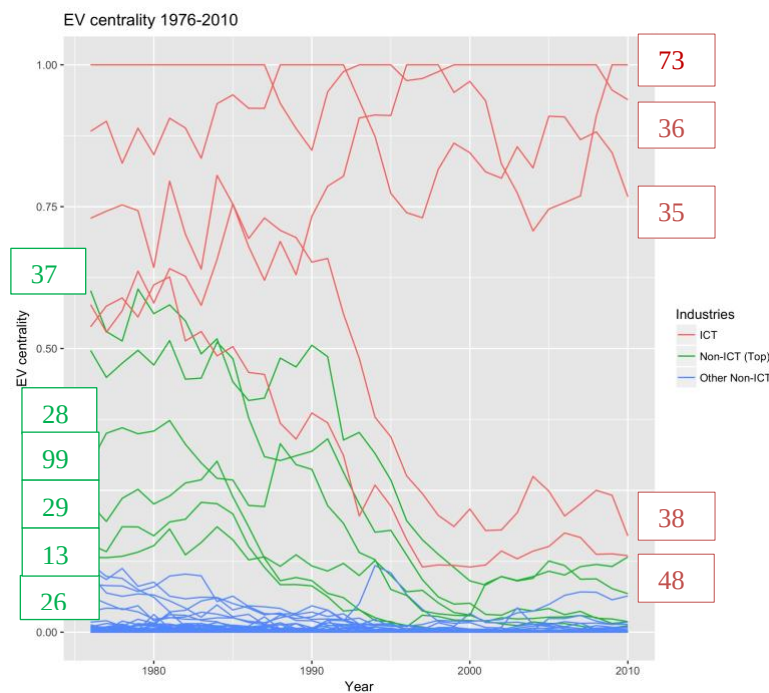


Figure 2: Eigenvector (EV) centralities (normalized to 0-1 scale within a year) of all industries for the period 1976 – 2010; industries are represented by their SIC codes at 2-digit level; orange represents ICT industries; green represents relatively more central non-ICT industries; the rest of the non-ICT industries are represented in blue.

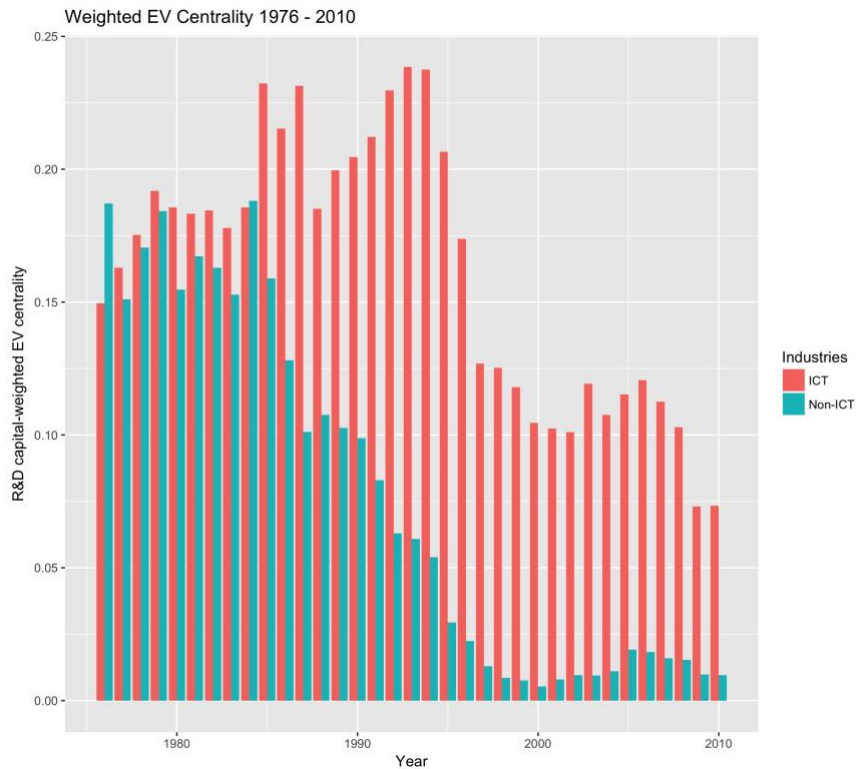


Figure 3: R&D capital-weighted eigenvector centralities of non-ICT (blue) and ICT (red) industries.

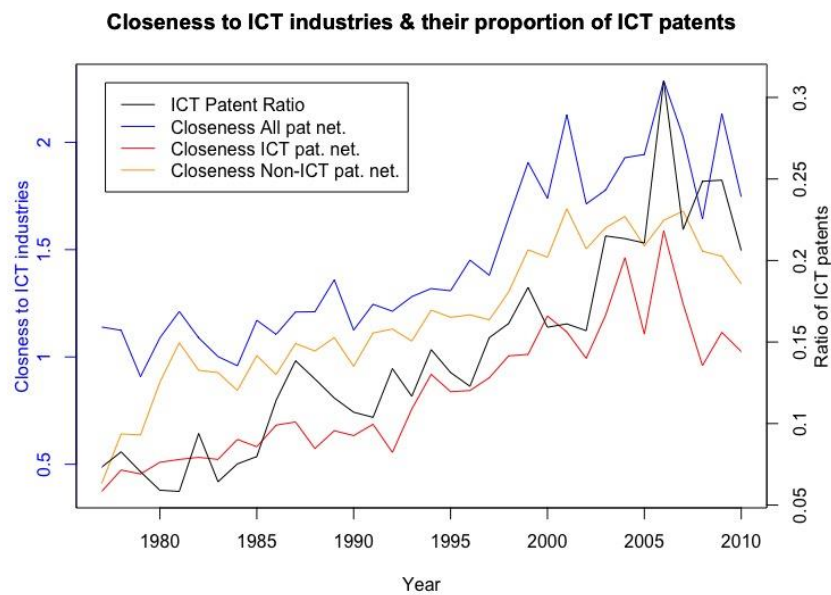


Figure 4: Red: ICT-Closeness of Non-ICT industries in the network constructed using ICT patents alone; Yellow: ICT-Closeness of Non-ICT industries in the network constructed using non-ICT patents alone; Blue: ICT-Closeness of Non-ICT industries in the network constructed using all patents; Black: Proportion of ICT patents of non-ICT industries.

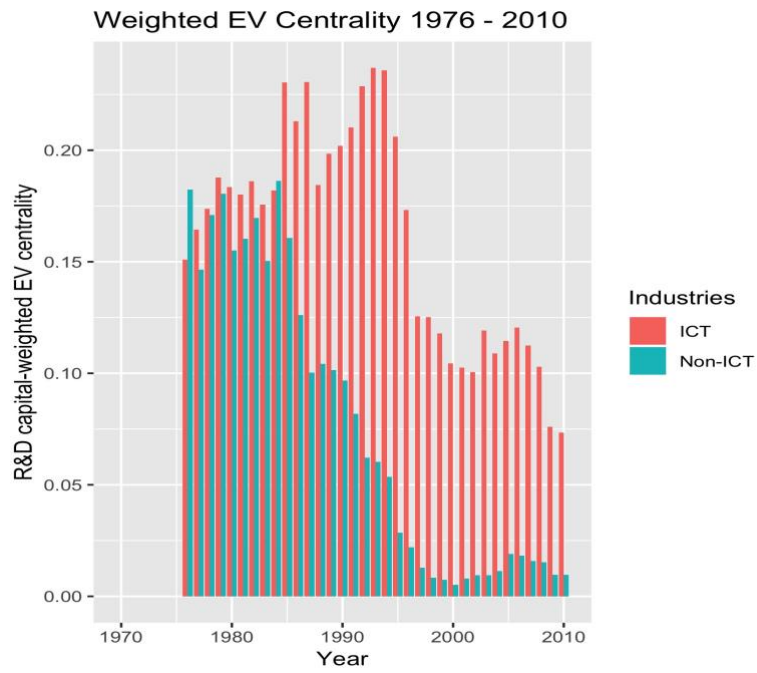


Figure 5: R&D Capital weighted eigenvector centralities of non-ICT (blue) and ICT (red) industries as per the simulated network using ERGM.

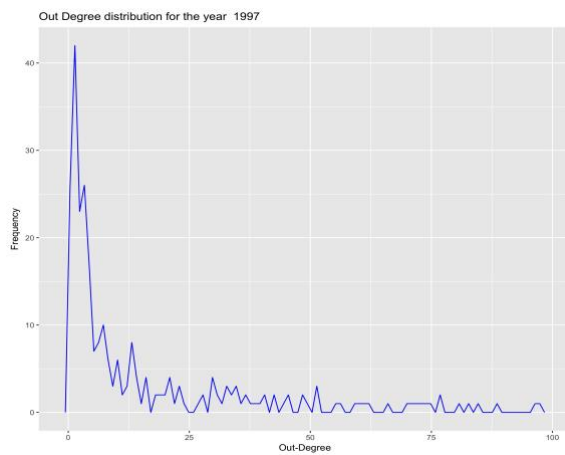


Figure 6a: Out-degree distribution of actual citation network in the year 1997

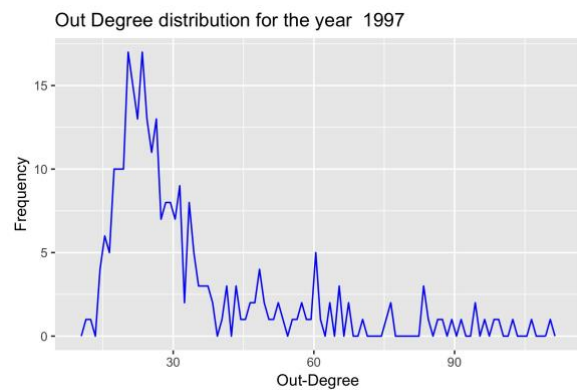


Figure 6b: Out-degree distribution of simulated citation network in the year 1997

Figure 6: Comparison of the degree distribution of the real and simulated citation networks - 1997

Tables

Table 1: Monte Carlo MLE results summary for the citation network corresponding to Year 1997 (1997 is taken as a representative year; results for other years have been omitted for reasons of brevity)

Year → Sufficient Statistics ↓	1997
Propensity to cite based on $Experience_{i,y}$	0.002** (0.001)
Propensity to get cited based on $Experience_{i,y}$	0.002** (0.0008)
AIC	-256389.210
BIC	-267324.6173

Table 2: Summary statistics of independent and dependent variables

<i>Variable</i>	<i>Observations</i>	<i>Mean</i>	<i>Standard Deviation</i>
<i>ln(IEff)</i>	8,132	1.036	1.793
<i>Reuse</i>	8,132	0.309	1.449
<i>Creation</i>	8,132	0.333	0.421
<i>CF</i>	8,132	0.030	0.201
<i>CF_Spread</i>	8,132	0.177	0.318
<i>CF_Vol</i>	8,132	0.015	0.048
<i>ICTCloseness</i>	8,132	0.980	1.263
<i>EVCentrality</i>	8,132	0.008	0.041
<i>MktInt</i>	8,132	0.457	4.690
<i>RnDInt</i>	8,132	0.474	8.963
<i>PatHHI</i>	8,132	0.469	0.439
<i>SaleHHI</i>	8,132	0.594	0.300
<i>AT</i>	8,132	5.813	2.011
<i>ME</i>	8,132	5.272	2.045
<i>BTM</i>	8,132	0.604	0.515
<i>ROA</i>	8,132	-0.212	10.727
<i>ITCapital</i>	8,132	0.007	0.024
<i>NoICTCapDummy</i>	8,132	0.176	0.381
<i>ln(BMPatents)</i>	8,132	0.299	0.642
<i>ICTPatRatio</i>	8,132	0.062	0.179
<i>Examiner Experience</i>	8,132	11.012	6.137

Table 3: Stage 2 output of the instrumented network approach — regressions of Innovation outcomes

Dependent variable → Independent variable ↓	(I) $\ln(IEff_{i,y})$	(II) $Reuse_{i,y}$	(III) $Creation_{i,y}$
$ICTCloseness_{i,y-1}$	0.102*** (0.029)	0.147*** (0.023)	0.032*** (0.007)
$EVCentrality_{i,y-1}$	0.095*** (0.021)	0.038** (0.017)	0.013** (0.005)
$MktInt_{i,y-1}$	-0.017 (0.016)	-0.001 (0.013)	0.002 (0.004)
$RnDInt_{i,y-1}$	0.003 (0.016)	-0.004 (0.013)	0.006 (0.004)
$PatHHI_{i,y-1}$	0.182*** (0.019)	0.011 (0.015)	0.029*** (0.004)
$SaleHHI_{i,y-1}$	-0.033 (0.027)	-0.016 (0.021)	-0.027*** (0.006)
$AT_{i,y-1}$	0.251*** (0.045)	0.049 (0.035)	0.020* (0.010)
$ME_{i,y-1}$	0.067* (0.035)	-0.035 (0.028)	0.005 (0.008)
$BTM_{i,y-1}$	0.055** (0.023)	0.006 (0.018)	0.002 (0.005)
$ROA_{i,y-1}$	-0.072*** (0.016)	0.001 (0.012)	-0.009** (0.004)
$ITCapital_{i,y-1}$	-0.001 (0.020)	-0.001 (0.016)	0.002 (0.005)
$NoICTCapDummy_{i,y-1}$	-0.008 (0.021)	-0.024 (0.016)	-0.002 (0.005)
$IEff_{i,y-1}$	0.366*** (0.026)		
$\partial(IEff_{i,y-1})$	-0.030* (0.016)		
$Reuse_{i,y-1}$		0.754*** (0.018)	
$\partial(Reuse_{i,y-1})$		-0.475*** (0.015)	
$Creation_{i,y-1}$			0.059*** (0.006)
$\partial(Creation_{i,y-1})$			-0.035*** (0.005)
Constant	0.363 (0.266)	5.052*** (0.236)	0.404*** (0.062)
Industry Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	8,132	8,132	8,132
Number of Industries	306	306	306
R-squared	0.414	0.442	0.428

Robust Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Stage 2 output of the instrumented network approach — regressions of industry competitiveness

Dependent variable → Independent variable ↓	(I) $CF_{i,y}$	(II) $CF_Spread_{i,y}$	(III) $CF_Vol_{i,y}$
$ICTCloseness_{i,y-1}$	-0.001 (0.003)	0.012*** (0.005)	0.001* (0.001)
$EVCentrality_{i,y-1}$	0.001 (0.002)	0.003 (0.004)	0.001 (0.0004)
$MktInt_{i,y-1}$	0.0004 (0.002)	0.001 (0.003)	-0.0001 (0.0003)
$RnDInt_{i,y-1}$	-0.001 (0.002)	0.0003 (0.003)	0.0002 (0.0003)
$PatHHI_{i,y-1}$	0.005** (0.002)	-0.002 (0.003)	0.0003 (0.0003)
$SaleHHI_{i,y-1}$	-0.002 (0.003)	-0.017*** (0.004)	0.0003 (0.0005)
$AT_{i,y-1}$	0.006 (0.005)	-0.012* (0.007)	-0.002*** (0.001)
$ME_{i,y-1}$	-0.0002 (0.004)	0.016*** (0.006)	0.001 (0.001)
$BTM_{i,y-1}$	0.001 (0.003)	0.001 (0.004)	0.002*** (0.0004)
$ROA_{i,y-1}$	0.0001 (0.002)	0.001 (0.003)	0.0001 (0.0003)
$ITCapital_{i,y-1}$	-0.001 (0.002)	-0.005 (0.003)	-0.0003 (0.0003)
$NoICTCapDummy_{i,y-1}$	0.001 (0.002)	0.006* (0.003)	-0.0002 (0.0004)
$CF_{i,y-1}$	0.054*** (0.003)		
$\partial(CF_{i,y-1})$	-0.001 (0.002)		
$CF_Spread_{i,y-1}$		0.097*** (0.004)	
$\partial(CF_Spread_{i,y-1})$		-0.033*** (0.003)	
$CF_Vol_{i,y-1}$			0.025*** (0.0004)
$\partial(CF_Vol_{i,y-1})$			0.004*** (0.0003)
$Constant$	0.008 (0.030)	-0.054 (0.044)	0.012** (0.005)
$Observations$	8,132	8,132	8,132
$Number\ of\ Industries$	306	306	306
$R-squared$	0.397	0.499	0.641
$Industry\ Fixed\ Effects$	Yes	Yes	Yes
$Year\ Fixed\ Effects$	Yes	Yes	Yes

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5: Mechanism Results (Stage 2 output of the Instrumented Network Approach) - Similarity, digitization, and business method patents

Dependent variable → Independent variable ↓	(I) $ICTPatRatio_{i,y}$	(II) $Pat_Sim_1yr_{i,y}$	(III) $Pat_Sim_5yr_{i,y}$	(IV) $ln(BMPatents_{i,y})$
$ICTCloseness_{i,y-1}$	0.016*** (0.004)	0.005*** (0.001)	0.003*** (0.001)	0.097*** (0.015)
$EVCentrality_{i,y-1}$	0.003 (0.002)	0.001** (0.002)	0.003*** (0.001)	0.015 (0.009)
$MktInt_{i,y-1}$	0.001 (0.001)	-0.001** (0.0002)	-0.001*** (0.0002)	-0.0003 (0.002)
$RnDInt_{i,y-1}$	-0.003* (0.002)	0.0002 (0.0002)	0.001** (0.0003)	-0.003 (0.003)
$PatHHI_{i,y-1}$	-0.002 (0.002)	0.001 (0.001)	0.001 (0.001)	-0.001 (0.004)
$SaleHHI_{i,y-1}$	-0.002 (0.004)	-0.006*** (0.001)	-0.005*** (0.001)	0.001 (0.008)
$AT_{i,y-1}$	-0.004 (0.006)	0.002 (0.002)	0.002 (0.002)	0.008 (0.012)
$ME_{i,y-1}$	0.005* (0.003)	0.005*** (0.002)	0.004*** (0.001)	0.003 (0.004)
$BTM_{i,y-1}$	-0.0002 (0.002)	0.002** (0.001)	0.002*** (0.001)	-0.007 (0.005)
$ROA_{i,y-1}$	-0.0001 (0.001)	-0.001*** (0.000)	-0.001*** (0.000)	-0.0003 (0.001)
$ITCapital_{i,y-1}$	0.005 (0.003)	0.001 (0.001)	0.001 (0.001)	0.004* (0.002)
$NoICTCapDummy_{i,y-1}$	0.004* (0.002)	-0.001 (0.001)	0.000 (0.001)	0.003 (0.006)
$ICTPatRatio_{i,y-1}$	0.058*** (0.007)			
$\partial(ICTPatRatio_{i,y-1})$	-0.021*** (0.005)			
$Pat_Sim_1yr_{i,y-1}$		0.0221*** (0.002)		
$\partial(Pat_Sim_1yr_{i,y-1})$		-0.009*** (0.001)		
$Pat_Sim_5yr_{i,y-1}$			0.019*** (0.002)	
$\partial(Pat_Sim_5yr_{i,y-1})$			-0.007*** (0.001)	
$ln(BMPatents_{i,y-1})$				0.374*** (0.032)
$\partial(ln(BMPatents_{i,y-1}))$				-0.078*** (0.011)
Constant	0.062*** (0.0002)	0.098*** (0.003)	0.033*** (0.003)	0.299*** (0.0003)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	8,131	8,132	8,132	8,131
Number of Industries	306	306	306	306
R-squared	0.364	0.108	0.183	0.761

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$